

Complete solutions to all problems are required. Models should be specified, approximations and conclusions should be specified and motivated.

Problems 1, 2 and 3 can give a maximum of 10 points each. Problems 4, 5 and 6 can give a maximum of 20 points each. The maximum total on the exam is 90 points. The limit for Pass (P) is at least 45 points, and the limit for High Pass (HP) is at least 67 points.

Only the paper sheets provided by the institute in the exam room shall be used, both for solutions and for sketches/notes.

Each problem solution shall start at the top of a new paper.

Write your ID number on each paper that you hand in.

Red pencil or pen is not allowed.

Calculators are allowed. Only statistical tables and formula sheets provided by the institute are allowed. No other tables or formula sheets are allowed.

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1. Let Y_1, Y_2, Y_3 be i.i.d. random variables which take values in $\{3, 4\}$, and with probability mass function $f(3) = c(\theta)\theta$, $f(4) = c(\theta)\theta^3$, where $c(\theta)$ is a function. The observations of Y_1, Y_2, Y_3 are

4, 4, 3.

The parameter space is $\Theta = \mathbf{R}_+$. Find the maximum likelihood estimator of θ . [10p]

2. Suppose that $x_1, \dots, x_7$ are (not necessarily independent) observations of the random variable X , and $y_1, \dots, y_4$ are (not necessarily independent) observations of the random variable Y . It is known that $E(X) = \theta$ and $E(Y) = 3\theta$, and that the parameter space is $\theta = \mathbf{R}_+$. The data are

x'_i : 1.73, 1.67, 1.84, 1.06, 1.80, 1.21, 1.30
 y'_i : 2.25, 3.41, 5.53, 6.76.

Find the least squares estimator of θ based on the data $x_1, \dots, x_7, y_1, \dots, y_4$. [10p]

3. We have one observation $u = 3.5$ of a random variable U which is $Exp(1/\theta^2)$ distributed, so that U has the exponential distribution with $E(U) = 1/\theta^2$. The parameter space is $\Theta = \mathbf{R}$. Construct a 95% twosided confidence interval for θ . [10p]

4. We have ten observations of the i.i.d. random variables $X_1, \dots, X_{10}$ which are $Bern(\theta)$ distributed. They are

0, 0, 0, 1, 1, 0, 1, 0, 1, 0.

(a) Perform an exact test of the hypotheses

$$\begin{aligned} H_0 : & \quad \theta \geq 4/5, \\ H_1 : & \quad \theta < 4/5. \end{aligned}$$

Can you reject the null hypothesis? [6p]

- (b) Suppose instead that the observations are 0, 0, 0, 1, 0, 0, 0, 0, 0, 0. Your friend, who is interested in mathematics, asks you which is the smallest value of θ_0 for which you could reject the null hypothesis in

$$\begin{aligned} H_0 : & \quad \theta \geq \theta_0, \\ H_1 : & \quad \theta < \theta_0. \end{aligned}$$

on level 0.05, when using an exact test as in (a). What do you answer your friend? (You may use numerical approximations if you are not able to derive the exact value for θ_0). [6p]

- (c) Suppose that data are as given in (a). Find the plug-in estimator of $P(X_1 + \dots + X_{10} = 1)$, based on the maximum likelihood estimator of θ . [8p]

5. We have one observation $x = 3$ from a random variable $X \in \text{Bin}(50, \theta)$.

- (a) Perform an exact test, on level 0.01, for the hypotheses

$$\begin{aligned} H_0 : & \quad \theta \geq 0.1, \\ H_1 : & \quad \theta < 0.1. \end{aligned}$$

Can the null hypothesis be rejected? [5p]

- (b) Construct an approximate lower bounded 95 % confidence interval for θ , using an appropriate approximation. [5p]

- (c) Construct an approximate level 0.01 test for the hypothesis in (a), and calculate the (approximate) power of the test at $\theta = 0.05$, using the approximation in (b). [10p]

6. Suppose that U is a continuous random variable with unknown distribution function F . We have the following 20 i.i.d. observations of U :

1.60, 4.44, 2.01, 0.563, 6.04, 0.0311, 7.04, 0.433, 1.12, 0.00520,
3.95, 2.08, 7.15, 0.0662, 1.31, 0.0354, 2.67, 1.96, 1.33, 1.11.

We are interested in estimating the parameter $\theta = E(U^3)$. The parameter space is $\theta = \mathbf{R}_+$.

- (a) Derive and calculate the plug-in estimator $\hat{\theta}_n$ of θ . [6p]
 (b) Show that $\hat{\theta}_n$ is unbiased. [6p]
 (c) Suppose that it is known that $E(U^6) < \infty$. Show that $\hat{\theta}_n$ is consistent in quadratic mean. [8p]